Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/26/2024
PI Name:	Miljkovic, Nenad Ph.D.		
Project Title:	High-Fidelity Experiments and Computations of Transient Two-Phase Flow for Understanding Cryogenic Propellant Tank Transfer		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Urbana	**State:**	IL
Zip Code:	61801-2924	**Congressional District:**	13
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2020 Physical Sciences NNH20ZDA012N: Fluid Physics. Appendix A
Start Date:	11/30/2021	**End Date:**	11/29/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:	No PI changes to report. The same postdoc and PhD student are continuing work on the project.		
COI Name (Institution):	Yazdani, Miad Ph.D. (United Technologies Corporation)		
Grant/Contract No.:	80NSSC22K0294		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Cryogenic chilldown governs initial stages of cryogen transport. Chilldown involves complicated hydrodynamic and thermal interactions between the liquid, vapor, and channel wall. Large initial temperature differences between the walls and the cryogen create rapid evaporation and large pressure and temperature fluctuation. Although work has attempted to identify transient flow boiling regimes, local surface temperatures, and heat fluxes, chilldown remains poorly understood due to the lack of experimental techniques capable of attaining high spatio-temporal resolution in both optical and infrared (IR) spectra. Furthermore, a lack of computational methods exists which can predict transient flow boiling and chilldown for a variety of length and time scales, and working fluid and system geometries. Here, we propose a collaborative effort between the University of Illinois, Urbana-Champaign (UIUC) and Raytheon Technologies (RTX) to develop fundamental understanding of chilldown using complementary high-fidelity experiments and computations. Internal flow pattern variations ranging from the film, transition, nucleate, to convection flow boiling using FC-72 and liquid nitrogen (LN2) in NASA relevant aluminum and stainless steel tubes, will be studied. We will use in-liquid endoscopy to study in-situ quench front propagation during FC-72 and LN2 flow boiling. The synchronous use of internal optical and external IR visualization will enable the gaining of a rigorous understanding of the thermal-fluidic behavior occurring in the near-wall region during chilldown and transient flow boiling. The obtained parameters, such as the quench front propagation rate and the temperature and heat flux distributions near the quench front, will then be used to validate high-fidelity computations. The computational framework at RTX leverages the established foundation that is capable of predicting the thermal and hydrodynamic behavior of multiphase flows in convective boiling and condensation regimes. The multiple scales associated with chilldown and two-phase flow boiling will be addressed through a combination of the previously developed Direct Numerical Simulation (DNS) approach for the nucleation near the wall, the Large Eddy Simulation (LES) formulation for the macroscopic transport in the core, and a novel coupling scheme for transporting the information across these scales. While the inherent transient phenomena such as solid condition, nucleation, and regime transition are an integral part of this framework, the model will be modified to allow transient boundary and operating conditions due to the operation of the tank during chilldown. The simulation will provide highly resolved information on the thermal and flow characteristics of two-phase cryogenic flow during the chilldown process and particularly the transient evolution of the flow regime during boiling. The model predictions will continually be validated against the high-fidelity experimental measurements over a range of test conditions. The work is broken down into tasks, which are briefly defined by: 1) Experimental analysis of transient heat transfer and pressure fluctuation during chilldown in Earth gravity with FC-72; 2) Integration of synchronous optical and IR visualization of chilldown with FC-72 and LN2 in Earth gravity for a variety of conditions including tube material, tube size, pressure difference, initial system temperature, and cryogen flow rate; 3) Simulations of transient flow boiling and chilldown in Earth gravity with FC-72 in order to provide physical insight on how the flow regime and boiling regime evolve over the course of chilldown; and 4) FC-72 flow boiling tests in microgravity with simulation validation. The outcomes of the on-Earth experiments will guide testing in microgravity on the Flow Boiling and Condensation Experiment (FBCE) on the International Space Station (ISS) to better understand the time-varying system pressure and temperatures during the cryogenic propellant transfer process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cryogenic chilldown governs initial stages of cryogen transport. Flow pattern variation and quench front propagation are crucial for analyzing and understanding the mechanism of chilldown.
Task Progress:	In the last reporting year, we conducted the chilldown experiments in the copper and stainless-steel tube with PP1 (FC-72) at different initial mass flow rates and subcoolings to investigate their effects on the heat transfer mechanisms and interface dynamics of transient flow boiling during the chilldown process. Therefore, in this reporting year, we comprehensively analyzed both the heat transfer data and transient system responses during the chilldown in the stainless-steel tube and completed the paper draft, which is now reviewed by the co-authors' team at University of Illinois, Urbana-Champaign (UIUC). In addition, we started synchronous flow visualization and infrared (IR) imaging for chilldown with FC-72 in Earth gravity early this year. We did many verification trials, finally optimized the heating strategy and figured out the non-uniformity issue of indium tin oxide (ITO) coating and the breaking issue of the visualization glass tubes. Even though we spent some time troubleshooting, we upgraded our test rig and acquired some excellent preliminary results about flow pattern transition and wall temperature evolution during the chilldown. This task will continue moving forward in the next reporting year. In addition, the work about FBCE-TL kept moving forward. In this reporting year, we had regular meetings with Case Western Reserve University (CWRU) and NASA Glenn Research Center (GRC) on the science requirement document (SRD). The science requirement document (SRD) successfully passed the science requirement review in January this year. Then, we kept revising and adjusting the statements and requirements in the integrated science requirements documents (ISRD) through discussion in bi-weekly meetings. Till now, the science requirements document has almost been well-organized and edited. The engineering team from GRC has already started the configuration design and manufacturing based on the updated ISRD.
Bibliography Type:	Description: (Last Updated: 11/30/2025)
Abstracts for Journals and Proceedings	Zhang J, Wang C, Garimella VS, Miljkovic N. "Perfluorohexane line chilldown in horizontal SS tubes during low Reynolds number flows." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Zhang J, Ganesan V, Garimella VS, Miljkovic N. "Spatio-temporal evolution of two-phase flow boiling regimes and heat transfer during line chilldown of nPFH in horizontal fused quartz tubes." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024